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Project: Power Bill: Residential Rates for Rockingham, Virginia

Company: Shenandoah Valley Electric Cooperative

Residential-A-13-June-2022.pdf (svec.coop)

Objectives:

1. Calculate the summer power bill of the residents of Rockingham, Virginia within each range of specific power usage manually using Arithmetic method.
2. Write a piece-wise function of the residential rates.
3. Recalculate the same power bill of the residents of Rockingham, Virginia within each range of specific power usage algebraically using the Piece-wise Function method.

All of this will be calculated using specifically Single-Phase usage for June-September billing cycle.

Information:

Type of Service: Single phase, 60-cycles at available voltage

Monthly Rate:

A. Distribution Charges:

1. Basic consumer charges:

   Single Phase @ $30.00 per billing month

B. Power Supply Charges:

1. June through September Billing:

First 800 kWh @ 6.380 cents per kWh

Over 800 kWh @ 9.024 cents per kWh

Numbers to be Tested:

0 kWh, 500 kWh, 1000 kWh, 1500 kWh

Arithmetic Method:

For the Arithmetic Method section, we'll be calculating the cost of the power bill by taking charge rates from the table in the Information section and multiplying them by the amount of power usage indicated by the values in the Numbers to be Tested section.

On the company website, we are given prices in cents/kWh.

$$6.380\ cents \div 100 = \$0.0638$$

$$9.024\ cents\ \div 100 = \$0.09024$$

These conversions from prices in cents to dollars were done so the answers we get will be in total dollars.

0 kWh

Base Charge = $30.00

Charge Rate = $0.0638/kWh

Actual Power Usage = 0 kWh

Cost = Base Charge + Charge Rate * Actual Power Usage

$$
\begin{aligned}
Cost &= \$30.00 + \$0.0638/kWh \cdot 0\ kWh \\
&= \$30.00 + \$0.0638/kWh \cdot 0\ kWh \\
&= \$30.00 + \$0.0638 \cdot 0 \\
&= \$30.00 + \$0.00 \\
&= \$30.00
\end{aligned}
$$

The cost for 0 kWh of power usage would be $30.00.

500 kWh

Base Charge = $30.00

Charge Rate = $0.0638/kWh

Actual Power Usage = 500 kWh

Cost = Base Charge + Charge Rate * Actual Power Usage

$$
\begin{aligned}
Cost &= \$30.00 + \$0.0638/kWh \cdot 500\ kWh \\
&= \$30.00 + \$0.0638 \cdot 500\ kWh \\
&= \$30.00 + \$0.0638 \cdot 500 \\
&= \$30.00 + \$31.90 \\
&= \$61.90
\end{aligned}
$$

The cost for 500 kWh of power usage is $61.90.

For the previous two calculations, the actual power usage has been less than or equal to 800 kWh. This means the same rate ($0.0638/kWh) has been used. For the following calculation, a power usage over 800 kWh, it will be necessary to calculate the cost up to 800 kWh as in previous calculations. However, it will then be necessary to use the second rate ($0.09024/kWh) for all power usage over 800 kWh. Since the total usage is 1000 kWh and the rate changes at 800 kWh, this will require subtracting 800 kWh from the total 1000 kWh, leaving 200 kWh. The remaining 200 kWh is the power usage over 800 kWh and will be calculated using the second, higher rate, and be added to the cost calculated for the usage up to 800 kWh.

1000 kWh

Base Charge = $30.00

Charge Rate = $.0638/kWh for the first 800 kWh plus $.09024/kWh for all kWh over 800

Actual Power Usage = 1000 kWh

Power Usage Up to 800 kWh = 800 kWh

Power Usage Over 800 kWh = 1000 kWh - 800 kWh = 200 kWh

Cost = Base Charge +

Charge Rate Up to 800 kWh * Power Usage Up to 800 kWh +

Charge Rate Over 800 kWh * Power Usage Over 800 kWh

$$Cost = \$30.00 +$$

$$\$0.0638/kWh \cdot 800\ kWh +$$

$$\$0.09024/kWh \cdot 200\ kWh$$

$$= \$30.00 +$$
$$\$0.0638\, kWh \cdot 800\, kWh +$$
$$\$0.09024\, kWh \cdot 200\, kWh$$

$$= \$30.00 + \$0.0638 \cdot 800 + \$0.09024 \cdot 200$$
$$= \$30.00 + \$51.04 + \$18.048$$
$$= \$81.04 + \$18.048$$
$$= \$99.088$$
$$\approx \$99.09$$

The cost for 1000 kWh of power usage is $99.09.

1500 kWh

$$Cost = \$30.00 +$$
$$\$0.0638/kWh \cdot 800\, kWh +$$
$$\$0.09024/kWh \cdot 700\, kWh$$

$$= \$30.00 +$$
$$\$0.0638\, kWh \cdot 800\, kWh +$$
$$\$0.09024\, kWh \cdot 700\, kWh$$

$$= \$30.00 + \$0.0638 \cdot 800 + \$0.09024 \cdot 700$$
$$= \$30.00 + \$51.04 + \$63.168$$
$$= \$81.04 + \$63.168$$
$$= \$144.208$$

$$\approx \$144.21$$

The cost for 1500 kWh of power usage is $144.21.

Piece-wise Function:

In the Arithmetic Method section, we established how to manually calculate the cost for a monthly power bill for a customer of Shenandoah Valley Electric Cooperative (SVEC). Here, we will develop a more concise, algebraic function called a piece-wise function in order to calculate the same values. In order to develop the function, we will first define the variables and equations necessary to define the problem. Instead of performing the calculations manually at this point, however, we will simplify the equations and combine them, as necessary, for concision. Lastly, we will express those simplified equations as a part of a single piece-wise function.

On the company website, we are given prices in cents/kWh.

$$6.380\ cents \div 100 = \$0.0638$$

$$9.024\ cents \div 100 = \$0.09024$$

These conversions from prices in cents to dollars were done so the answers we get will be in total dollars.

The total cost, C, for a monthly power bill for a SVEC residential customer is given by the equation:

$$C = b + p,$$

where b is the base charge and p is the power usage charge. According to the table in the Information section, the base charge, b, is a constant $30.00.

So, the first piece of our piece-wise function is:

$$C(p) = \{\$30.00 + \$0.0638p,\ if\ 0 \leq x \leq 800$$

The second piece of our piece-wise function will be formed based on the knowledge that once the number of kilowatts goes over 800, the rate changes to $0.09024/kWh.

The total cost if usage goes over 800 kWh is given by the equation:

$$C(p) = \$30.00 + 0.0638(800) + 0.09024(p - 800)$$

$$C(p) = \$30.00 + 51.04 + 0.09024p - 72.192$$

$$C(p) = 81.04 - 72.192 + 0.09024p$$

$$C(p) = 8.848 + 0.09024p$$

So, the second piece of our piece-wise function will be:

$$C(p) = \{0.09024p + 8.848,\ if\ p > 800$$

Finally, our fully developed piece-wise function becomes:

$$C(p) = \begin{cases} 0.0638p + 30.00, & if\ 0 \leq p \leq 800 \\ 0.09024p + 8.848, & if\ p > 800 \end{cases}$$

Piece-Wise Function Method:

0 kWh

0 falls into the first piece of our piece-wise function so our equation will be as follows:

$$C(p) = 0.0638(0) + 30$$

$$C(p) = \$30.00$$

The total cost for 0 kWh of power usage is $30.00.

500 kWh

500 falls into the first piece of our piece-wise function so our equation will be as follows:

$$C(p) = 0.0638(500) + 30$$

$$C(p) = 31.90 + 30.00$$

$$C(p) = \$61.90$$

The total cost for 500 kWh of power usage is $61.90.

1000 kWh

1000 falls into the second piece of our piece-wise function so our equation will be as follows:

$$C(p) = 0.09024(1000) + 8.848$$

$$C(p) = 90.24 + 8.848$$

$$C(p) = 99.088$$

$$C(p) \approx 99.09$$

The total cost for 1000 kWh of power usage is $99.09.

1500 kWh

1500 falls into the second piece of our piece-wise function so our equation will be as follows:

$$C(p) = 0.09024(1500) + 8.848$$

$$C(p) = 135.36 + 8.848$$

$$C(p) = 144.208$$

$$C(p) \approx 144.21$$

The total cost of 1500 kWh of power usage is $144.21.

Citations (APA):

Shenandoah Valley Electric Cooperative. Residential-A-13-June-2022.pdf. Retrieved June 6, 2023, from https://www.svec.coop/wp-content/uploads/Residential-A-13-June-2022.pdf.

Chukwuemeka, S. Piecewise functions. Piecewise Functions. Retrieved June 6, 2023, from https://piecewise-functions.appspot.com/#studentProjectPowerBill